



ASSAM POWER DISTRIBUTION COMPANY LIMITED.
OFFICE OF THE CHIEF GENERAL MANAGER (D & S),
(LOWER ASSAM REGION)
CIN: U40109AS2003SGC007242

No CGM (D&S)/APDCL/LAR/Bongaigaon Medical College/2025-26/ 68

Date: 09/07/2026

To,

M/s U.K. Enterprise,
R.P. Road, Dispur

Sub: **Approval of GTP and drawings.**

Ref: 1. Letter received from CGM(PP&D) vide No. CGM(PP&D)/APDCL/GTPA/PART-23/2025-26/4735 dated 07/07/2026.
2. Letter received from M/s U.K. Enterprise, R.P. Road, Dispur vide letter no. UKE/BMC/2026-27/157 dated: 02.07.2026

With reference to the above subject, this is to inform you that GTP along with drawings of item mentioned below is hereby approved subject to the strict compliance to the highlighted corrections and recommendations of the GTP. All factory accepted tests shall be conducted upon the item and submitted along with supply and verified before installation. The date of manufacturing of the said item should be after the date of issue of this letter.

➔ GTP for the following items have been approved.

Sl. No.	Description of item	Vendor Details
1	33KV XLPE Cable, 1X630Sqmm	M/s Havells India Alwar, Rajasthan, India
2	11KV XLPE Cable, 1X630Sqmm	
3	Copper control cable 10 core, 2.5sqmm	
4	Copper control cable 4 core, 2.5sqmm	

- N.B: 1. *It is to be noted that this approval is solely against the furnished GTP of the said material and shall not be treated as approval of vendor under APDCL.*
2. *All Tests Certificates are to be submitted along with the supply.*

Encl: - Approved GTPs.

Yours faithfully

Chief General Manager (D&S)

Memo No CGM (D&S)/APDCL/LAR/Bongaigaon Medical College/2025-26/

Date:

Copy to:

1. The CGM(PP&D), APDCL, Bijulee Bhawan, Paltanbazar, Ghy-01 for your information.
2. The CEO, Bongaigaon-EC, APDCL, LAR for information and necessary action.
3. Relevant File

Chief General Manager (D&S)

674

অসম শক্তি বিতৰণ কোম্পানী লিমিটেড
ASSAM POWER DISTRIBUTION COMPANY LIMITED
[CIN / চিআইন- U40109AS2003SGC007242]

No : CGM(PP&D)/APDCL/GTPA/PART-23/2025-26/4735

Dated : 07/07/26

To,

Chief General Manager (LAR)
5th Floor, Bijulee Bhawan
Paltan bazar, Guwahati-1.

Sub: Approval of GTP

Ref: i) Letter No, CGM(D&S)/APDCL/LAR/Bongaigaon Medical College/2025-2026/63, Dtd:03.07.26

Sir,

With reference to the subject cited above, it is to inform you that the GTP along with drawings submitted from your end vide under Ref: i) have been scrutinized by the technical team of the office of the undersigned and the following items are hereby approved subject to full compliance to the highlighted corrections and recommendations. Details of materials:

Sl No.	Item Description	Manufacture
1	33KV XLPE Cable, 1X630sqmm	M/S Havells India, Alwar, Rajasthan
2	11KV XLPE Cable, 1X630sqmm	
3	1.1KV, 10CX2.5sqmm Control Cable (Copper)	
4	1.1KV, 4CX2.5sqmm Control Cable (Copper)	

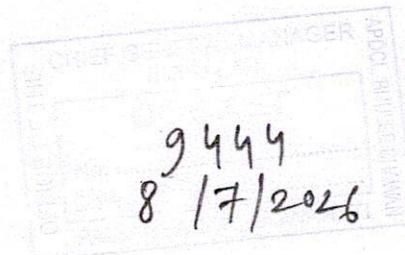
It is further clarified that M/s Havells India Ltd., Alwar, Rajasthan is not an approved vendor of APDCL for supply of 33 kV XLPE Cable, 1C x 630 sq.mm. However, based on the type test reports submitted, approval of the item at Sl. No. 1 is accorded on a one-time basis.

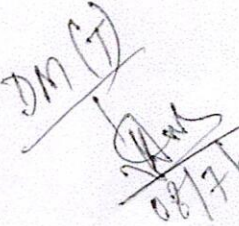
Moreover, you are also requested to ensure that the approved item supplied strictly comply with the approved GTP & highlighted corrections/recommendations. All factory acceptance tests shall be conducted upon the item. The date of manufacture of the said item shall be later than the date of issue of this letter.

This is for favour of your kind information and necessary action.

Yours sincerely


Chief General Manager (PP&D)




CGM (D&S)
APDCL (LAR)



ASSAM POWER DISTRIBUTION COMPANY LIMITED.

OFFICE OF THE CHIEF GENERAL MANAGER (D),
(LOWER ASSAM REGION)

CIN: U40109AS2003SGC007242

No. CGM (D&S)/APDCL/LAR/Bongaigaon Medical College/2025-26/

Date: 03/07/2026

To,

The Chief General Manager (PP&D),
APDCL, Bijulee Bhawan, 7
Paltan Bazar, Guwahati-1.

Sub: Forwarding of GTP & Drawing

Ref: 1. Letter received from M/s U.K. Enterprise, R.P.Road, Dispur, vide No. UKE/BMC/2026-27/157 dated: 02.07.2026.

Sir,

With reference to the above subject, enclosed please find herewith a letter received from M/s U.K. Enterprise, R.P.Road, Dispur regarding submission of GTP and drawings of following materials for the work Construction of 33KV Dedicated Medical College Feeder from 132KV/33KV Dhaligaon GSS to the premises of Bongaigaon Medical College, Majgaon and associated works under Bongaigaon-ESD-I against LoA issued vide No. CGM(D&S)/APDCL/LAR/ Bongaigaon Medical College /2025-2026/38 dated: 13.05.2026 for your information and necessary action.

Details are as follows :-

Sl. No.	Description of item	Vendor Details
1	33KV XLPE Cable, 1X630Sqmm	M/s Havells India Alwar, Rajasthan, India
2	11KV XLPE Cable, 1X630Sqmm.	
3	Copper control cable 10 core, 2.5sqmm	
4	Copper control cable 4 core, 2.5sqmm	

Encl:- 1. Copy of letter under Reference.
2. GTP and Drawing

Yours faithfully



Chief General Manager (D&S)
APDCL, LAR

M/S. U.K. ENT**RISE**

Govt. Approved (Class I) Contractor ,

*Deals with all Electrical, Civil works & Consultant***UDYAM REGISTRATION NO. UDYAM-AS-03-0005077**

Dr. R.P. Road, Dispur, Guwahati-781006, Ph.98643-46349, 90853-90161 Email: babultalukdar@gmail.com

No. UKE/ BMC/2026-27/157

Date 02.07.2026

To
✓ The Chief General Manager(D&S)
APDCL (LAR), Bijulee Bhawan ,
Paltan Bazar , Guwahati -1

Sub:- Approval of GTP and Drawing

Ref:- 1. LOA No. CGM(D&S)/APDCL/LAR/Bongaigaon Medical College /2025-26/38
Dt. 13.05.2026

Sir,

With reference to the above , we have the honour to forward herewith the GTP and Drawing of the following materials for favour of according necessary approval from your end .

The details are as follows :-

Sl. No.	Item Description	Name of Manufacturer	Remarks
1	33 KV XLPE Cable , 1cr. 630 sq. mm	M/S. Havells India Limited , Alwar, Rajasthan, India	
2	11 KV XLPE Cable , 1cr. 630 sq. mm		
3	Copper control Cable 10cr. 2.5 sq. mm		
4	Copper control Cable 4cr. 2.5 sq. mm		

The above materials will be utilised for Construction of 33 KV Dedicated Medical College Feeder from 132/33kV Dhaligaon GSS to the premises of the Bongaigaon Medical College, Majgaon and associated work under Bongaigaon ESD-I under Deposit Work Scheme.

Thanking you in anticipation and assuring our best service in future .

Encl. : As above

Yours faithfully
Authorised SignatoryM/S. U.K. Enterprise,
Dispur, Guwahati-069254
21/7/2026

Chowdhury, DMF

CGM (D&S)
APDCL (LAR)

NAME OF THE CONTRACTOR: M/s. U.K. Enterprise
NAME OF THE PROJECT: Bongaigaon Medical College
NAME OF THE CLIENT: Assam Power Distribution Company Limited
LOA NO.: CGM(D&M)/APDCL/LAR/Bongaigaon Medical College/2025-26/38 Dt. 13.05.26
NAME OF THE MANUFACTURER: HAVELLS INDIA LIMITED
CABLE DETAILS: 1x630
VOLTAGE GRADE 33KV E

HAVELLS INDIA LTD. (CABLE DIVISION) - ALWAR

HAVELLS

Technical Particulars

ID: 43616

Date: 05.06.2026

Sr.No.	Descriptions	Unit	TCX630
A General Detail			
1	SAP Code		
2	Nos. of Core		CMHRLDBH801630
3	Nominal cross sectional area	sq.mm	1.00
4	Type of Cable (Cable Code)		930.20
5	Applicable Standard		A2NFAY
6	Voltage Grade	kV	IS: 7096 PART-2 2011, IS: 8130:2013, IS: 8821:84 With up to date amendment
B Conductor			
1	Material		19/33
2	Shape of Conductor		ALUMINIUM E. C. GRADE
3	Nominal size		Standard compacted Circular
a) Main			
b) Reduced Neutral	sq.mm		550.00
4	Minimum No. of Strand in each Conductor		0.00
a) Main			
b) Reduced Neutral	sq.mm		03
			0
Note: Size of strand shall be corresponding to meet the requirements of conductor Resistance as per relevant clause of IS: 8130 with upto date amendment			
C Screening on Conductor			
1	Type & Material		
2	Thickness Approx	mm	Semiconducting Compound by extrusion
3	Continuous working temperature	deg.C	0.30
D Insulation			
1	Type & Material		90.00
2	Thickness Nominal		Extruded X.L.P.E. conforming to IS: 7096 PT-2
a) Main			
b) Reduced Neutral	mm		8.50
E Screening on Insulation (Non-Metallic Part)			
1	Type & Material		0.00
2	Thickness Approx		Semiconducting Compound by extrusion
F Copper Taping/Metallic Part			
1	Type & Material		0.30
2	Approx. thickness		Copper tape shall be wrapped helically with 100% coverage
3	Cores Identification	mm	0.040
4	Frictions calculated Laid up dia		NA
G Inner Sheath			
1	Type & Material		48.00
2	Application		Extruded PVC Type ST-2, conforming to IS: 5831
3	Thickness Minimum		BY-Extruded
H Filler			
1	Material		0.50
I Armour			
1	Type & Material		NA
2	Nom. Dim. of Armour Flat Strip / Round Wire		Single Layer of Aluminium Flat Strip, Conforming to IS: 7096 PT-2
J Outer Sheath			
1	Type & Material		4X0.89 = 19%
2	Application		
3	Thickness Nominal (For unarmoured cables)		Extruded PVC Type ST-2, conforming to IS: 5831
4	Thickness Minimum (For armoured/unarmoured cables)	mm	BY-Extruded
5	Colour of outer sheath		6.00
			2.20
			BLACK

"APPROVED"

CGM (T&C Circle)
PP&D, APDCL

CGM (T&C Circle)
PP&D, APDCL



K Special Properties of outer sheath			
1	Oxygen index test		
2	Temperature Index Test		
3	Acid Gas Generation		
4	Smoke generation Test		
5	Flammability test		
L Electrical Characteristics			
1	Max. Conductor DC Resistance at 20°C		
a)	Main	Ω/KM	0.0469
b)	Reduced Neutral	Ω/KM	0.0600
2	Approx. Conductor AC Resistance at max. Operating Temp.		
a)	Main	Ω/KM	0.0648
b)	Reduced Neutral	Ω/KM	0.0800
3	Approx. Resistance at 50 Hz	Ω/KM	0.0000
4	Approx. Capacitance of Cable	PF/KM	0.1110
5	Min. Volume resistivity of insulation	MΩ.CM	0.2900
6	Cont. Current Rating for Standard IS Condition when Laid		1 X 10° 24 at 27° Deg. C. & 1 X 10° 42 at 99 Deg. C
a)	In Ground at 30°C	AMPS	
b)	In Duct at 30°C	AMPS	495.00
c)	In Air at 49°C	AMPS	424.00
7	Allowable max. temp. When carrying Cont. rated current	DEG.C	77.00
8	Short Circuit Current Rating for 1 Sec. Duration	KVA	90.00
9	Max. Conductor Temperature allowed for the short circuit condition	DEG.C	50.00
10			250.00
M General Information			
1	Overall Diameter of cable (approx.)	MM	
2	Net weight of cable (approx.)	KG/KM	29.61
3	Standard drum length	Meter	4100.00
4	Net Standard drum length		500 - 5%
5	Recommended min. Bending radius of cable		Max. 5 % of ordered quantity
6	Recommended Max. Safe pulling force	N/A	20 X D (WHERE D IS OVERALL DIAMETER OF CABLE IN MM)
a)	When cond. pulled by pulling eye		
b)	When cable pulled by stacking		30 x T (where T is total cross sectional area of cond. in sq.mm)
7	Marking Detail (by embossing & printing)		R X D/2 (where D is overall diameter of cable in mm)
8	Sequential marking of length		HAVELLS 19.33KV XLPE ELECTRIC CABLE (SIZE IN sq.mm) (YEAR OF MFG.)
9	Packing Material		Shall be printed at every meter of length on outer sheath
			Non Returnable wooden drums

* Values indicated as "Approx." are for guidelines only

* In case of any Typographical Error / Discrepancy, relevant standards shall be prevailed.

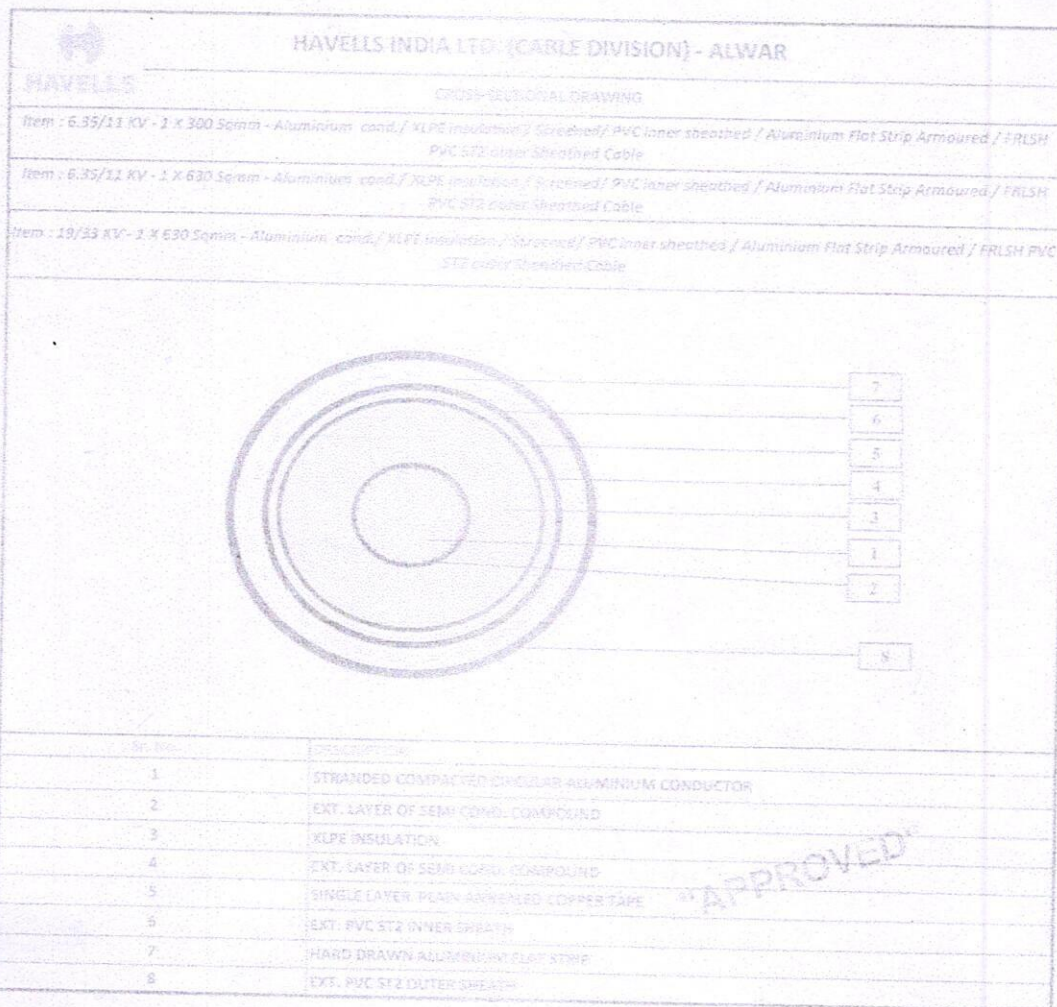
⑧ All test certificates to be submitted along with supply

APPROVED

DM (T&C Circle)
PP&D, APDCL

DM (T&C Circle)
PP&D, APDCL





DM (T&C) OP
PP&D, APDOL



PP&D, APDOL



NAME OF THE CONTRACTOR: M/s. U.K. Enterprise

NAME OF THE PROJECT: Bongaigaon Medical College

NAME OF THE CLIENT: Assam Power Distribution Company Limited

LOA NO.: CGM(D&M)/APDCL/LAR/Bongaigaon Medical College/2025-26/38 Dt. 23.05.25

NAME OF THE MANUFACTURER: HAVELLS INDIA LIMITED

CABLE DETAILS: 1X630

VOLTAGE GRADE 11KV E

HAVELLS INDIA LTD. (CABLE DIVISION) - ALWAR

HAVELLS

Technical Particulars

Our Ref. No.: HIL/GTP/LA/APDCL A116/2025, Dated-04.12.2025

HV 6.25/11 KV XLPE INSULATED ALUMINIUM POWER CABLE

Sr. No.	Descriptions	Unit	1CKN300	1CK630
1	Name of Manufacturer		HAVELLS INDIA LIMITED	
2	Applicable Standard to which cable conforms		IS: 7096 Part-2, 2011 With up to date amendment	
3	Type of cable		A2XWbY	
4	Voltage Grade	kV	11 KV (E)	
5	Conductor detail			
	Conductor material		Aluminium H4 Grade	
	Nominal cross-sectional Area	mm ²	630	630
	Min Number of strands in each conductor		33	33
	Shape of conductor		Stranded circular compacted as per class-2 of IS 8130/2013	
6	Screening on Conductor:			
	Type & Material		Semiconducting Compound by extrusion	
	Approx. Thickness	mm	0.50	0.50
	Continuous working temperature	Deg. C	90	
7	Insulation:			
	Material		XLPE conforming to IS: 7096 Pt. 2 /2011	
	Nominal thickness	mm	3.60	3.60
8	Screening on Insulation:			
	Non metallic part		Black extruded Semiconducting compound (Stripable by adequate Heat - Bonded Type)	
	Material Type			
	Approx. thickness	mm	0.40	0.40
	metallic part			
	Material Type		Approx. 33 strands constitute the metallic part of Screening	
9	Identification of cores		Not applicable	
10	Filler material		Not applicable	
11	Inner Sheath:		Not applicable	
12	Armouring:		Not applicable	
	Material		Single layer of Hard Drawn Aluminium Round Wire	
	Type		Conforming to IS : 7096 Pt-2	
	Nominal diameter of wire	mm	2.00	
	Thickness of armour		More than 2 KA/3 Sec	
13	Outer Sheath:			
	Material		PRISM P.V.C.	
	Type		Type ST-2 conforming to IS : 5631/84	
	Thickness (Min)	mm	1.85	1.85
	Colour of outer sheath		BLUE	
14	Approximate overall diameter	mm	45	
15	Net weight of cable (approx.)	Kg/Km	3050	
16	Standard drum length	Mtr	500 ± 5%	
17	Non Standard drum length		Max. 5 % of ordered quantity	
	Overall Quality Tolerance		± 2%	
18	Continuous Current Rating for Standard IS Condition when laid:			
	Under ground at 30°C	Amps	476	
	In ducts	Amps	290	
	In air at 40°C	Amps	725	
	Allowable max. cond. Temp. when carrying rated current	Deg. C	90	
19	Short Circuit Rating			
	Short Circuit current rating for 1 Sec duration	kA/Sec	59.22	
	Allowable max. cond. Temp. at the time of short circuit	Deg. C	250	

CGM (T&C) (dgr)
PP&D, APDCL



20	Electrical parameters			
	Conductor DC Resistance at 20°C, (Max.)	Ohm/km	0.107	0.0466
	Conductor AC Resistance at 30°C (approx.)	Ohm/km	0.130	0.0543
	Approx. Resistance of cable at 50Kc	Ohm/km	0.113	0.104
	Approx. capacitance of cable	micro Farad	0.26	0.66
	Insulation Resistance, Volume resistivity, (Min)			
	at 27 Deg.C	Ohm-cm		1×10^{15}
	at 50 Deg.C	Ohm-cm		1×10^{12}
21	Marking Detail (by embossing/printing)	HAVELLS 0.35/11KV XLPE ELECTRIC CABLE (SIZE IN SQ. MM) (Year of mfg.)-FRISH- APDCL (ADB Project)		
22	Sequential marking of length	Shall be printed on every meter of length on outer sheath		
23	Packing Material	Non Returnable wooden/Steel drums (as per final Drums schedule)		
	FRISH PROPERTY :-			
	~ Oxygen index value at room temp.	23 % (Min.) As per ASTM-D 2863		
	~ Temp. index at 21% oxygen index value	250 Deg.C (Min.) As per ASTM-D 2863		
	~ Smoke density rating	E2 % (Max) As per ASTM-D 2243		
	~ Acid gas generation	20 % (Max) by weight, As per IEC -754-1		
	~ Flammability test	As per IEC-332-1		
* Values indicated as " Approx. " are for guidelines only.				
* In case of any Typographical Error / Discrepancy, relevant standards shall be prevailed.				

All test certificates to be submitted along with supply

Am
DM (T&C Circle)
PR&D, APDCL

[Signature]



[Signature]
DM (T&C Circle)
PR&D, APDCL





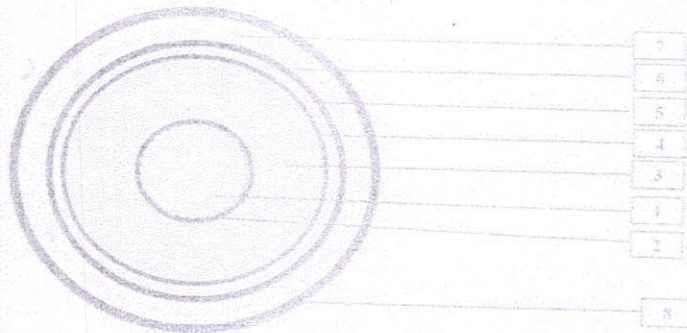
HAVELLS INDIA LTD. (CABLE DIVISION) - ALWAR

CROSS-SECTIONAL DRAWING

Item : 6.35/11 KV - 1 X 300 Sqmm - Aluminium cond./XLPE insulation / Screened / PVC inner sheathed / Aluminium Flat Strip Armoured / FRLSH PVC ST2 outer Sheathed Cable

Item : 6.35/11 KV - 1 X 630 Sqmm - Aluminium cond./XLPE insulation / Screened / PVC inner sheathed / Aluminium Flat Strip Armoured / FRLSH PVC ST2 outer Sheathed Cable

Item : 19/33 KV - 1 X 630 Sqmm - Aluminium cond./XLPE insulation / Screened / PVC inner sheathed / Aluminium Flat Strip Armoured / FRLSH PVC ST2 outer Sheathed Cable



Sr. No.	DESCRIPTION
1	STRANDED COMPACTED CIRCULAR ALUMINIUM CONDUCTOR
2	EXT. LAYER OF SEMI COND. COMPOUND
3	XLPE INSULATION
4	EXT. LAYER OF SEMI COND. COMPOUND
5	SINGLE LAYER PLAIN ANNEALED COPPER TAPE
6	EXT. PVC ST2 INNER SHEATH
7	HARD DRAWN ALUMINIUM FLAT STRIP
8	EXT. PVC ST2 OUTER SHEATH

DM (Cable)
PPSD APDCL

DM (Cable)
PPSD APDCL



NAME OF THE CONTRACTOR: M/s. U.K. Enterprise

NAME OF THE PROJECT: Bongaigaon Medical College

NAME OF THE CLIENT: Assam Power Distribution Company Limited

LOA NO.: CGM(D&M)/APDCL/LAR/Bongaigaon Medical College/2025-26/38 Dt. 13.05.26

NAME OF THE MANUFACTURER: HAVELLS INDIA LIMITED

CABLE DETAILS: 10x2.5

VOLTAGE GRADE 1.1KV

HAVELLS INDIA LTD. (CABLE DIVISION) - ALWAR

Technical Particulars

ID: 43616

Date: 30.06.2026

Sr.No.	Descriptions	Unit	10CX2.5
A General Detail			
1	SAP Code		CHCAMLDPHX10CX2.5
2	No. of Core	PCS	10.00
3	Nominal Cross sectional area	CM ²	2.50
4	Type of Cable/Cable Code		2XCV
5	Applicable Standard		IS: 7098 PART-I & IS: 8130-30; IS: 1975-99; IS: 5931-84 With up-to-date amendment.
6	Voltage Grade	KV	1.1
B Conductor			
1	Material		PLAIN ANNEALED COPPER
2	Shape of Conductor		Stranded Circular
3	Nominal area		
4	Main	CM ²	2.50
5	Reduced Neutral		0.00
6	Minimum Nom. Strands in each Conductor		
7	Main	PCS	1
8	Reduced Neutral		0
Str. Size of strand shall be corresponding to one of the requirements of conductivity Resistance as per relevant clause of IS: 8130 with up-to-date amendment.			
C Screening on Conductor			
1	Type & Material		NA
2	Thickness Approx.		0.00
3	Continuous working temperature		0.00
D Insulation			
1	Type & Material		XLPV, conforming to IS: 1048 P1.4
2	Thickness Nominal		
3	Main	MM	0.70
4	Reduced Neutral	MM	0.10
E Screening on Insulation(Non-Metallic Part)			
1	Type & Material		NA
2	Thickness Approx.		0.00
F Copper Taping/Metallic Part			
1	Type & Material		NA
2	Approx. thickness		0.00
3	Core identification		IS: 7098 PART-I
4	Dimensions calculated End up dia	MM	13.31
G Inner Sheath			
1	Type & Material		Extruded PVC, Test ST-2, conforming to IS: 5833
2	Application		BY-Extruded
3	Thickness Minimum	MM	0.30
H Filler			
1	Material		PVC
I Armour			
1	Type & Material		Single Layer of Galvanized steel Flat Strip, Conforming to IS: 5833
2	Nom. Dim. of Armour Flat Strip - Round Wire	MM	4X0.8 ± 10%

CLASSIFICATION: PUBLIC

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DM (T&C Circle)
PP&D, APDCL

DM (T&C Circle)
PP&D, APDCL



1	Armour		
2	Outer Sheath		
1	Type & Material		Extruded PVC Type SV 2, conforming to IS 1554
2	Application		BY F grade 3
3	Thickness Nominal (For minimum tensile test)	MM	0.05
4	Thickness Minimum (For minimum tensile test)	MM	1.40
5	Colour of outer sheath		BLACK
K	Special Properties of outer sheath		
1	Oxygen index test		
2	Temperature Index Test		
3	Acid Gas Generation		
4	Smoke generation Test		
5	Flammability test		
L	Electrical Characteristics		
1	Max. Conductor DC Resistance at 20°C		
a)	Main	OHMS/KM	0.0100
b)	Reduced Neutral	OHMS/KM	0.0050
2	Approx. Conductor AC Resistance at max. Operating Temp		
a)	Main	OHMS/KM	0.0100
b)	Reduced Neutral	OHMS/KM	0.0050
3	Approx. Reactance at 50 Hz	OHMS/KM	0.0000
4	Approx. Capacitance of Cable	PF/KV-KM	0.0000
5	Min. Volume resistivity of insulation	OHMS-CM	1 X 10 ¹⁴ at 27 Deg. C & 1 X 10 ¹² at 90 Deg. C
6	Cond. Current Rating for Standard IS Conditions when Tied		
a)	In Ground at 30°C	AMPERE	25.00
b)	In Duct at 30°C	AMPERE	25.00
c)	In Air at 40°C	AMPERE	25.00
7	Allowable max. cond. Temp. When carrying Cont. rated current	DEGREE C	90.00
8	Short Circuit Current Rating for 1 Sec. Duration	KVA/CM	0.36
9	Max. Conductor Temperature allowed for the short circuit condition	DEGREE C	250.00
M	General Information		
1	Overall Diameter of cable (approx.)	MM	26.00
2	Net weight of cable (approx.)	KG/KM	690.00
3	Standard drum length	MM	1000 (500 ± 5%)
4	Non Standard drum length		Max. 7% of ordered quantity
5	Recommended min. Bending radius of cable	MM	12 D WHERE D IS OVERALL DIAMETER OF CABLE IN MM
6	Recommended Max. Safe pulling force		
a)	When cond. pulled by pulling eye	N	30 X T where T is total cross sectional area of cond. in sq.mm
b)	When cable pulled by racking	N	5 X D ² where D is overall diameter of cable in mm
7	Marking Detail (by embossing / printing)		HAVELLS 100% XLPE ELECTRIC CABLE (SIZE IN sq.mm)
8	Sequential marking of length		Shall be printed at every meter of length on outer sheath
9	Packing Material		Non Returnable wooden drums

* Values indicated as "Approx." are for guidelines only

* In case of any Typographical Error / Discrepancy, relevant standards shall be prevailed.

① All tests certificates to be submitted

AM
DM (SG Grade)
PF&D, APDCL

CLASSIFICATION | PUBLIC

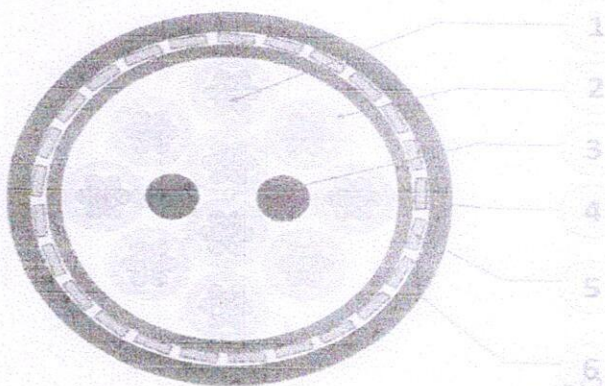




HAVELLS INDIA LTD. (CABLE DIVISION) - ALWAR

CROSS-SECTIONAL DRAWING

HV CABLE:- 10C X 2.5 SQ.MM 1100 V - Annealed Copper Copper Conductor / Extruded XLPE Insulation / Extruded PVC Inner Sheath / Galvanised Steel Flat Strip Armoured / Extruded PVC Outer Sheathed Cable As per IS 7098-1/1988.



APPROVED

DWG NOT TO SCALE

Sr. No.	DESCRIPTION
1	PLAIN ANNEALED COPPER - (STRANDED CIRCULAR-CLASS 2)
2	EXT. XLPE INSULATION
3	EXT. PVC ST-2 INNER SHEATH
4	GALVANISED STEEL FLAT STRIP
5	EXT. PVC ST-2 OUTER SHEATH

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PP&D, APDCCL

Signature

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PP&D, APDCCL



NAME OF THE CONTRACTOR: M/s. U.K. Enterprise

NAME OF THE PROJECT: Bongaigaon Medical College

NAME OF THE CLIENT: Assam Power Distribution Company Limited

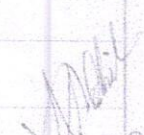
LOA NO.: CGM(D&M)/APDCL/LAR/Bongaigaon Medical College/2025-26/38 Dt. 13.05.26

NAME OF THE MANUFACTURER: HAVELLS INDIA LIMITED

CABLE DETAILS: 4x2.5

VOLTAGE GRADE 1.1KV

HAVELLS		HAVELLS INDIA LTD. (CABLE DIVISION) - ALWAR	
Technical Particulars			
Our Ref. No. HIL/GTP/EA/APDCL AIB/2025, Dated -04.12.2025		LV 1.1KV PVC INSULATED COPPER CABLE	
Descriptions	Unit	4CX2.5	
1. Name of Manufacturer.		HAVELLS INDIA LIMITED	
2. Type of cable(Cable code)		YWY	
3. Applicable Standard.		IS : 1554 Part-I, 1988 IN GENERAL	
4. Voltage Grade.	VOLTS	1100	
5. CONSTRUCTIONAL DETAIL			
a) Conductor :			
~ Material		Annealed Tinned Copper	
~ Nominal Size	mm ²	2.5	
		Min. size shall be corresponding to meet the requirements of conductor Resistance as per relevant clause of IS:-8130/2013	
~ Minimum Number of wires in each conductor	No.	3	
~ Shape of conductor		Stranded circular as per class -2 of IS 8130/2013	
b) Insulation :			
~ Material		HR PVC Type C	
~ Type		Conforming to IS -5831	
~ Nominal Thickness	mm	0.90	
~ Cores identification		Red, Yellow, Blue, Black & Grey upto 5 cores & more than 5 Cores, Grey Colour with Black number printing	
c) Inner Sheath :			
~ Material		FRLSH PVC	
~ Type		Type ST-2 conforming to IS -5831	
~ Application		BY - EXTRUSION	
~ Minimum Thickness	mm	0.30	
d) Armour :			
~ Material		Galvanized steel round wires	
~ Type		Conforming to IS-3975	
~ Nominal dimensions of armour wires	mm	1.40	
e) Outer Sheath :			
~ Material		FRLSH PVC	
~ Type		Type ST-2 conforming to IS -5831	
~ Application		BY - EXTRUSION	
~ Minimum Thickness	mm	1.24	
~ Color		BLACK	
6. ELECTRICAL CHARACTERISTICS			
a) Max. Conductor DC Resistance at 20°C	Ohm/km	7.56	
b) Max. Conductor AC Resistance at Max. Operating temperature	Ohm/km	9.53	
c) Approx. Reactance at 50 Hz.	Ohm/km	0.107	
d) Approx. Capacitance of cable (Core to core)	pF/km	0.22	
e) Minimum Volume resistivity of insulation	Ohm-cm	1X10 ¹⁴ at 27 Deg. C. & 1X10 ¹² at 85 Deg. C	
f) Continuous Current Rating for Standard IS Condition when laid,			
~ In Ground At 30°C	Amps	30	
~ In Duct	Amps	22	
~ In Air At 40°C	Amps	28	
g) Allowable max. Conductor temp. When carrying rated current	Deg. C	85	


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iv)	Short Circuit Rating for 1 Sec. Duration	1A	0.260
v)	Conductor Temperature allowed for the short circuit condition (Max)	°C	160
PACKING DETAIL			
	Standard drum length	mm	1000 / 500 ± 5%
	Non Standard drum length		Max. 5 % of ordered quantity
	Overall Quantity Tolerance		± 2%
7	GENERAL		
	~ Overall diameter of cable (Approx.)	mm	16
	~ Net weight of cable (approx.)	kg/km	520
	~ Recommended min. Bending radius of cable	mm	12 x D (where D is overall diameter of cable in mm)
	~ Recommended Max. Safe pulling force		
	i) When cond. Pulled by pulling eye	N	50 x T; where T is total crosssectional area of cond. in sqmm
	ii) When cable pulled by stocking	N	9 x D ² (where D is overall diameter of cable in mm)
	~ Sequential marking of length		Shall be printed at every meter of length on outer sheath
	~ Marking Detail (by embossing/printing)		HAVELIS 1100V PVC ELECTRIC CABLE (SIZE IN SQ.MM)-HR 85 - FRLSH - APDCL (ARB Project).
	~ Packing Material		Non Returnable wooden drums
FRLSH PROPERTY :-			
	~ Oxygen index value at room temp.		29 % (Min.) As per ASTM-D 2863
	~ Temp. index at 21% oxygen index value		120 Deg.C (Min.) As per ASTM-D 2863
	~ Smoke density rating		50 % (Max.) As per ASTM-D 2543
	~ Acid gas generation		20 % (Max.) by weight, As per IEC -754-1
	~ Flammability test		As per IEC-332-1
* Values indicated as " Approx. " are for guidelines only.			
* In case of any Typographical Error / Discrepancy, relevant standards shall be prevailed.			

⑧ All tests certificates to be submitted along with supply

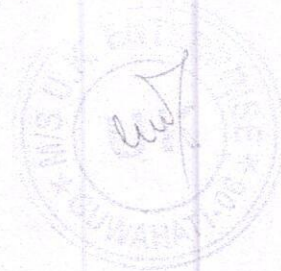
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HAVELLS INDIA LTD. (CABLE DIVISION) - ALWAR

CROSS-SECTIONAL DRAWING

Item : LV 1.1 KV 4CX2.5 2XWY - Copper conductor / XLPE insulation / PVC inner sheathed / G.S. wire armoured / PVC Sheathed Cable As per IS 7058 - 1/1993



- 5
- 4
- 3
- 2
- 1

S.No	Description
1	CONDUCTOR - PLAIN ANNEALED COPPER OF CLASS - 2 AS PER IS - 8130 (Stranded Circular)
2	INSULATION - XLPE
3	INNER SHEATH - EXTRUDED PVC ST2
4	ARMOUR - GALVANIZED STEEL ROUND WIRE
5	OUTER SHEATH - EXTRUDED PVC ST2

APPROVED



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PP&D, APDCL

